Absolute Maximum Ratings Symbol	Conditions [1]	Values	Units
V_{CES}		1200	V
V_{CGR}	R_{GE} = 20 kΩ	1200	V
I_C	T_{case} = 25/80 °C	260 / 180	A
I_{CM}	T_{case} = 25/80 °C; t_p = 1 ms	520 / 360	A
V_{GES}		± 20	V
P_{tot}	per IGBT, T_{case} = 25 °C	1040	W
T_j, (T_{stg})		–40 ... +150 (125)	°C
V_{isol}	AC, 1 min.	2500	V
humidity	IEC 60721-3-3	class 3K7/IE32	
climate	IEC 68 T.1	40/125/56	
Inverse Diode and FWD of type „GAL“ [6] [8]			
$I_F = -I_C$	T_{case} = 25/80 °C	200 / 160	A
$I_{FM} = -I_{CM}$	T_{case} = 25/80 °C; t_p = 1 ms	520 / 360	A
I_{FSM}	t_p = 10 ms; sin.; T_j = 150 °C	1450	A
I^2t	t_p = 10 ms; T_j = 150 °C	10 500	A^2s

Characteristics Symbol	Conditions [1]	min.	typ.	max.	Units
$V_{(BR)CES}$	V_{GE} = 0, I_C = 4 mA	≥ V_{CES}	–	–	V
$V_{GE(th)}$	V_{GE} = V_{CE}, I_C = 6 mA	4,5	5,5	6,5	V
I_{CES}	V_{GE} = 0, T_j = 25 °C	–	0,2	0,4	mA
	V_{CE} = V_{CES}, T_j = 125 °C	–	12	–	mA
I_{GES}	V_{GE} = 20 V, V_{CE} = 0	–	–	0,3	µA
V_{CEsat}	I_C = 150 A; V_{GE} = 15 V; T_j = 25 (125) °C	–	2,1(2,4)	2,45(2,85)	V
V_{CEsat}	I_C = 200 A; V_{GE} = 15 V; T_j = 25 (125) °C	–	2,5(3,0)	–	V
g_{fs}	V_{CE} = 20 V, I_C = 150 A	62	–	–	S
C_{CHC}	per IGBT	–	–	350	pF
C_{ies}	V_{GE} = 0	–	11	15	nF
C_{oes}	V_{CE} = 25 V	–	1,6	2	nF
C_{res}	f = 1 MHz	–	0,8	1	nF
L_{CE}		–	–	25	nH
$t_{d(on)}$	V_{CC} = 600 V	–	70	–	ns
t_r	V_{GE} = +15 V / -15 V [3]	–	55	–	ns
$t_{d(off)}$	I_C = 150 A, ind. load	–	490	–	ns
t_f	R_{Gon} = R_{Goff} = 7Ω	–	65	–	ns
E_{on}	T_j = 125 °C	–	26	–	mWs
E_{off}		–	23	–	mWs
Inverse Diode and FWD of type „GAL“ [6] [8]					
$V_F = V_{EC}$	I_F = 150 A; V_{GE} = 0 V; T_j = 25 (125) °C	–	2,0(1,8)	2,5	V
$V_F = V_{EC}$	I_F = 200 A; V_{GE} = 0 V; T_j = 25 (125) °C	–	2,25(2,05)	–	V
V_{TO}	T_j = 125 °C [2]	–	1,1	1,2	V
r_t	T_j = 125 °C [2]	–	–	7	mΩ
I_{RRM}	I_F = 150 A; T_j = 125 °C [2]	–	87	–	A
Q_{rr}	I_F = 150 A; T_j = 125 °C [2]	–	19	–	µC
Thermal characteristics					
R_{thjc}	per IGBT	–	–	0,12	°C/W
R_{thjc}	per diode	–	–	0,23	°C/W
R_{thch}	per module	–	–	0,05	°C/W

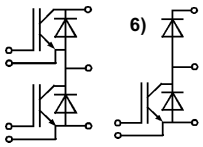
[1] T_{case} = 25 °C, unless otherwise specified
[2] I_F = – I_C, V_R = 600 V,
–di_F/dt = 1500 A/µs, V_{GE} = 0 V
[3] Use V_{GEoff} = –5... –15 V
[6] The free-wheeling diodes of the GAL type have the data of the inverse diodes.
[8] CAL = Controlled Axial Lifetime Technology

SEMITRANS® M
Low Loss IGBT Modules

SKM 195 GB 124 DN
SKM 195 GAL 124 DN



SEMITRANS 2N (low inductance)



GB **GAL**

Features

- N channel, homogeneous Silicon structure NPT-IGBT (Non punch through)
- Low saturation voltage
- Low inductance case
- Low tail current with low temperature dependence
- High short circuit capability, self limiting to 6 * I_{cnom}
- Fast & soft inverse CAL diodes [8]
- Without hard mould
- Large clearance (10 mm) and creepage distances (20 mm)

Typical Applications

- Switching (not for linear use)
- Switched mode power supplies
- DC servo and robot drives
- Inverters
- DC choppers
- AC motor speed control
- UPS Uninterruptable power supplies
- General power switching applications
- Electronic (also portable) welders

This datasheet has been downloaded from http://www.digchip.com at this page

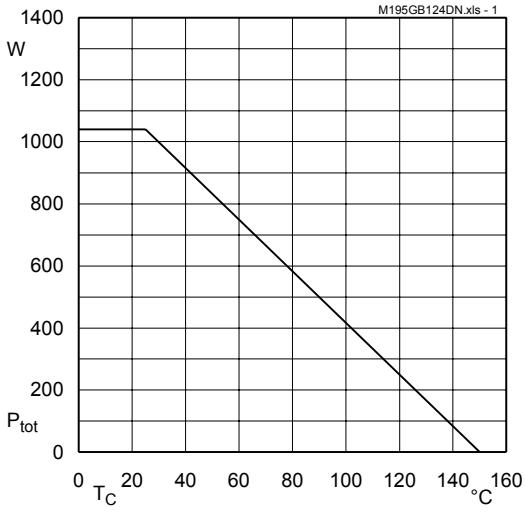


Fig. 1 Rated power dissipation $P_{tot} = f(T_C)$

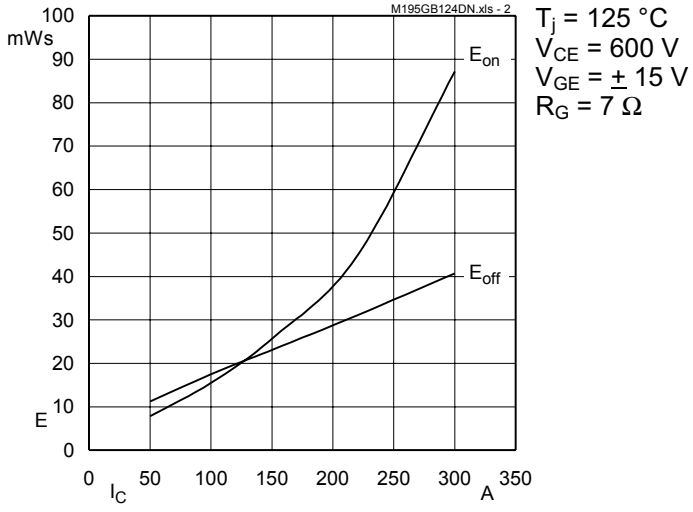


Fig. 2 Turn-on /-off energy = f (I_C)

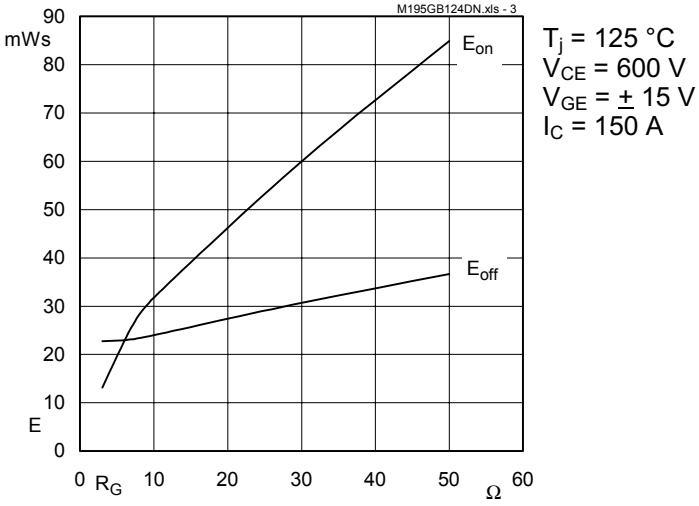


Fig. 3 Turn-on /-off energy = f (R_G)

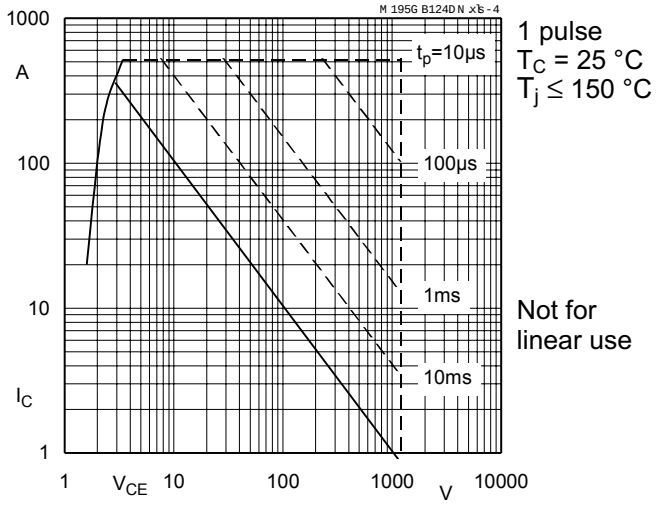


Fig. 4 Maximum safe operating area (SOA) $I_C = f(V_{CE})$

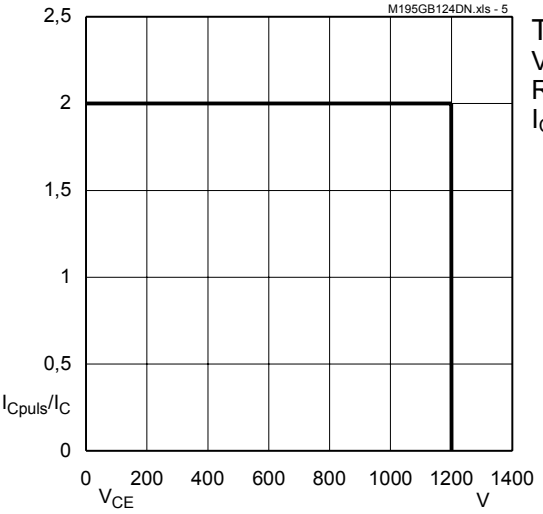


Fig. 5 Turn-off safe operating area (RBSOA)

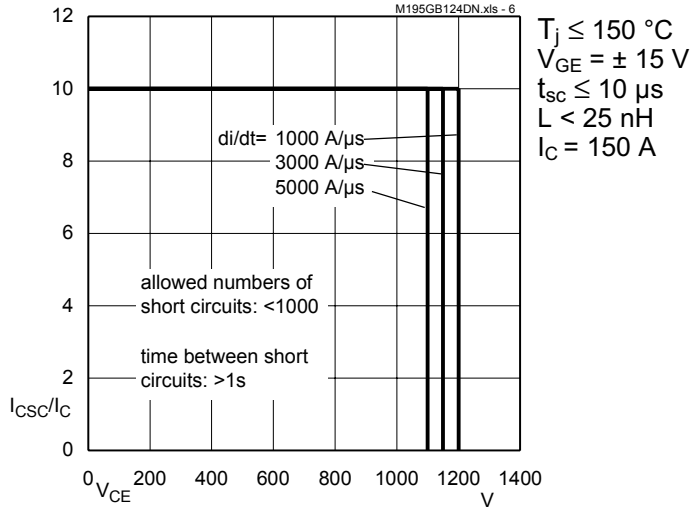
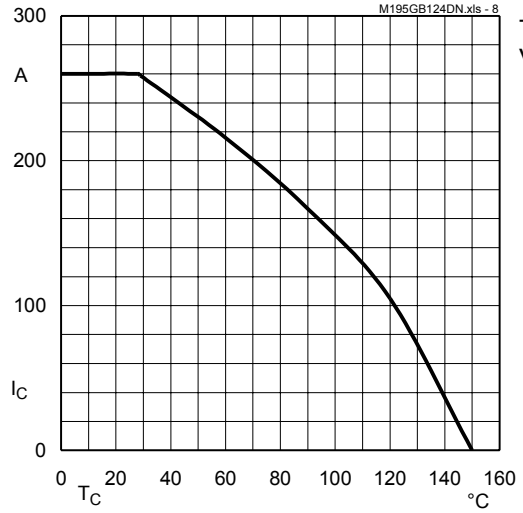


Fig. 6 Safe operating area at short circuit $I_C = f(V_{CE})$

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T_j = 150 °C
$V_{GE} \geq$ 15V

Fig. 8 Rated current vs. temperature $I_C = f(T_C)$

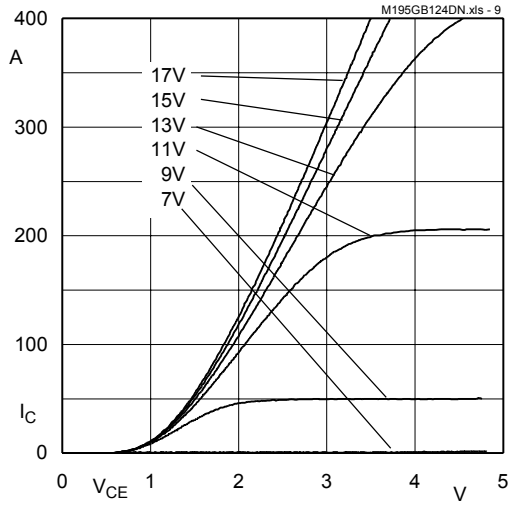


Fig. 9 Typ. output characteristic, t_p = 80 µs; 25 °C

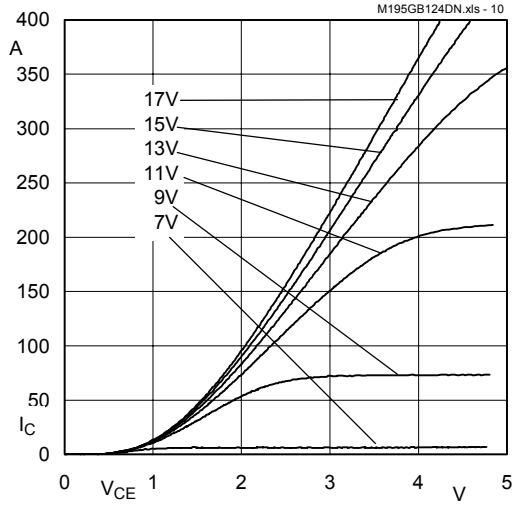


Fig. 10 Typ. output characteristic, t_p = 80 µs; 125 °C

$P_{cond(t)} = V_{CEsat(t)} \cdot I_{C(t)}$

$V_{CEsat(t)} = V_{CE(TO)(Tj)} + r_{CE(Tj)} \cdot I_{C(t)}$

$V_{CE(TO)(Tj)} \leq 1{,}29 + 0{,}0001\ (T_j - 25)$ [V]

typ.: $r_{CE(Tj)} = 0{,}0061 + 0{,}000022\ (T_j - 25)$ [Ω]

max.: $r_{CE(Tj)} = 0{,}0077 + 0{,}000022\ (T_j - 25)$ [Ω]

valid for $V_{GE} = +15^{+2}_{-1}$ [V]; $I_C > 0{,}3\ I_{Cnom}$

Fig. 11 Saturation characteristic (IGBT)
Calculation elements and equations

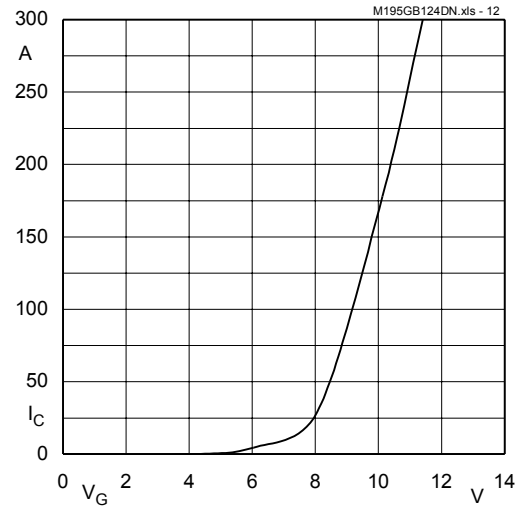
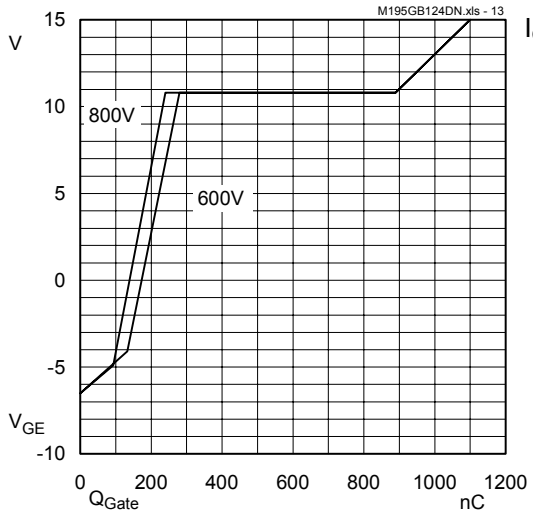
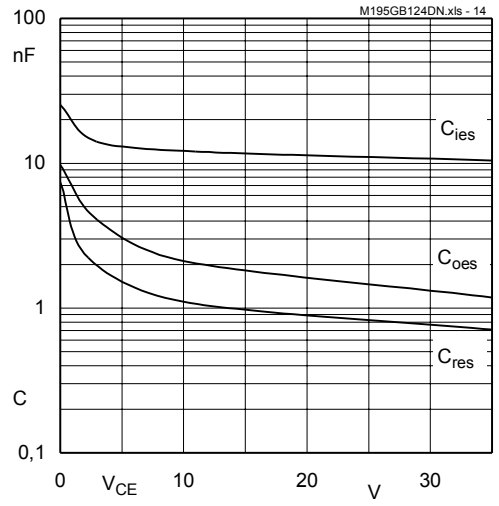


Fig. 12 Typ. transfer characteristic, t_p = 80 µs; V_{CE} = 20 V



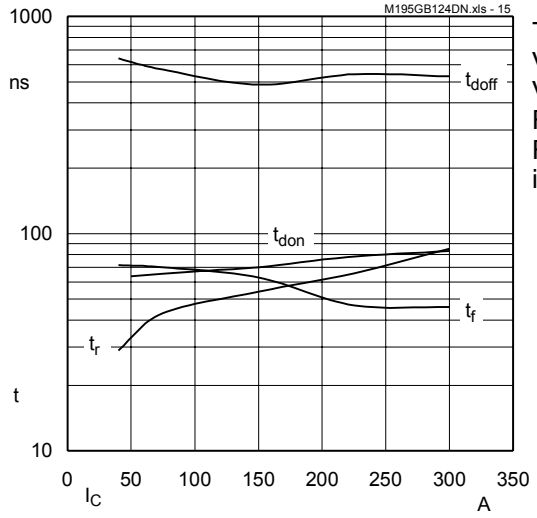
I_{Cpuls} = 150 A

Fig. 13 Typ. gate charge characteristic



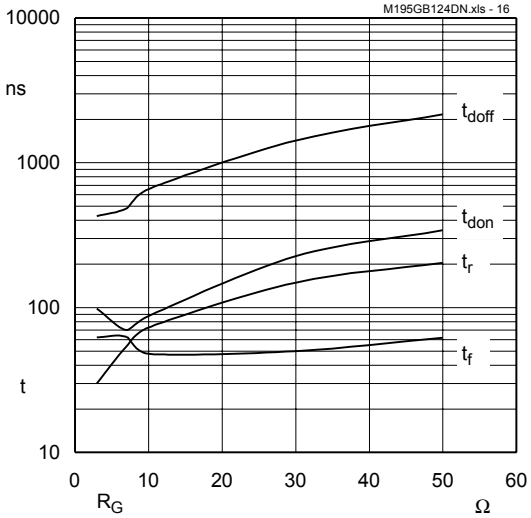
V_{GE} = 0 V
f = 1 MHz

Fig. 14 Typ. capacitances vs.V_{CE}



T_j = 125 °C
V_{CE} = 600 V
V_{GE} = ± 15 V
R_{Gon} = 7 Ω
R_{Goff} = 7 Ω
induct. load

Fig. 15 Typ. switching times vs. I_C



T_j = 125 °C
V_{CE} = 600 V
V_{GE} = ± 15 V
I_C = 150 A
induct. load

Fig. 16 Typ. switching times vs. gate resistor R_G

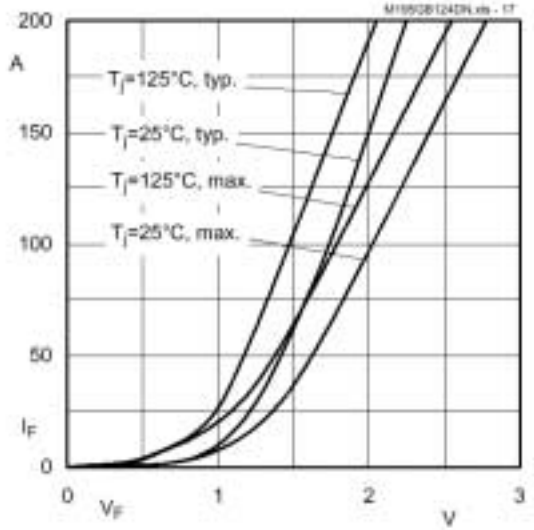
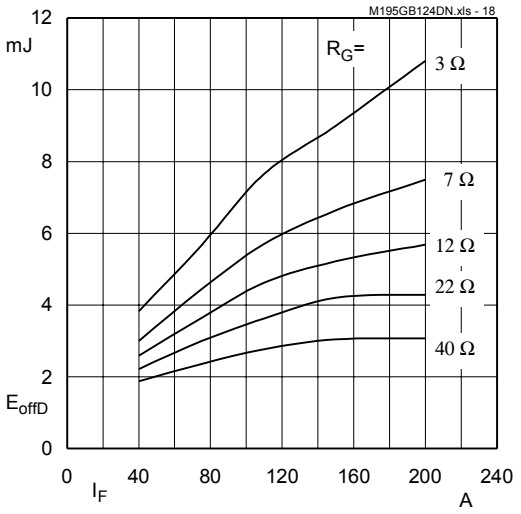


Fig. 17 Typ. CAL diode forward characteristic



V_{CC} = 600 V
T_j = 125 °C
V_{GE} = ± 15 V

Fig. 18 Diode turn-off energy dissipation per pulse

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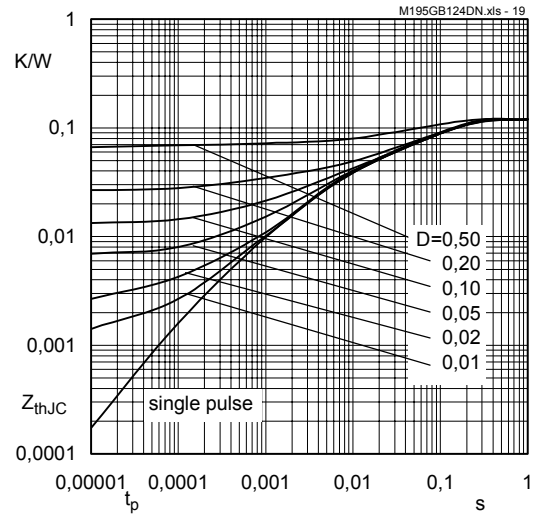


Fig. 19 Transient thermal impedance of IGBT
$Z_{thJC} = f(t_p)$; $D = t_p / t_c = t_p \cdot f$

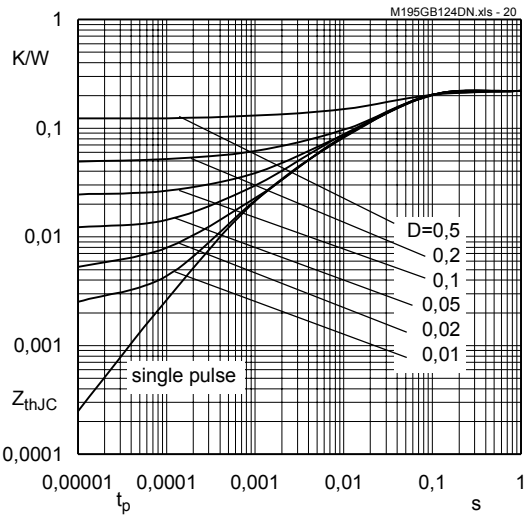


Fig. 20 Transient thermal impedance of inverse CAL diodes $Z_{thJC} = f(t_p)$; $D = t_p / t_c = t_p \cdot f$

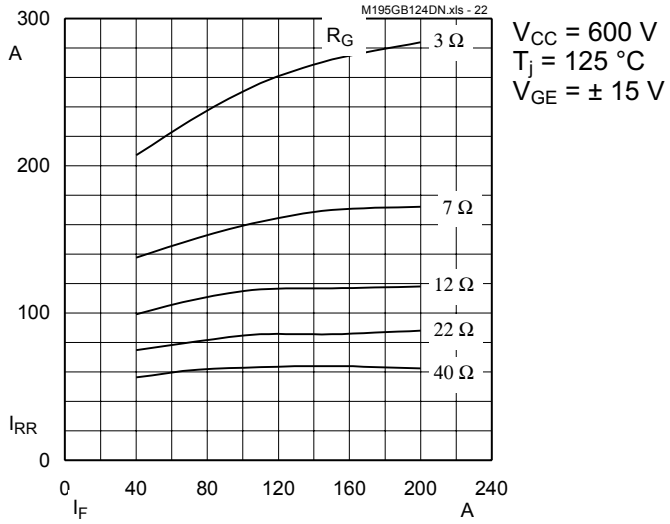


Fig. 22 Typ. CAL diode peak reverse recovery current $I_{RR} = f(I_F; R_G)$

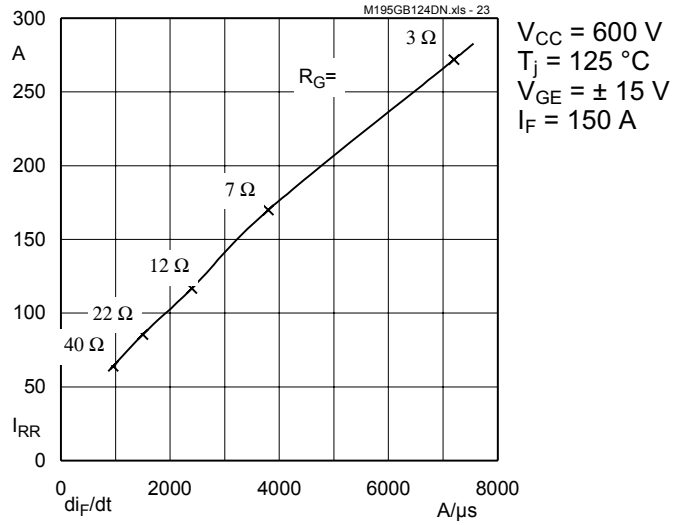


Fig. 23 Typ. CAL diode peak reverse recovery current $I_{RR} = f(di/dt)$

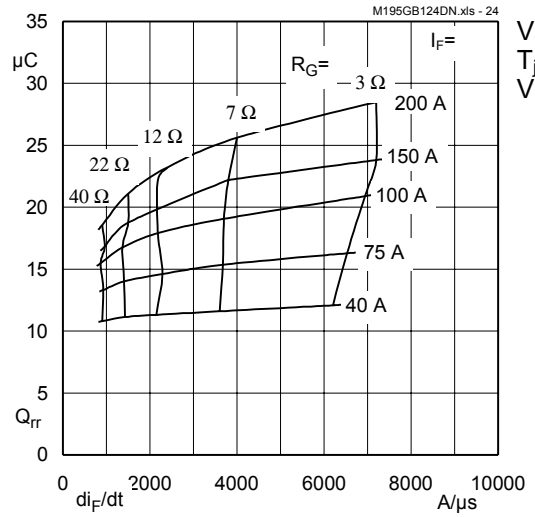
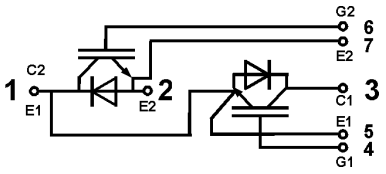


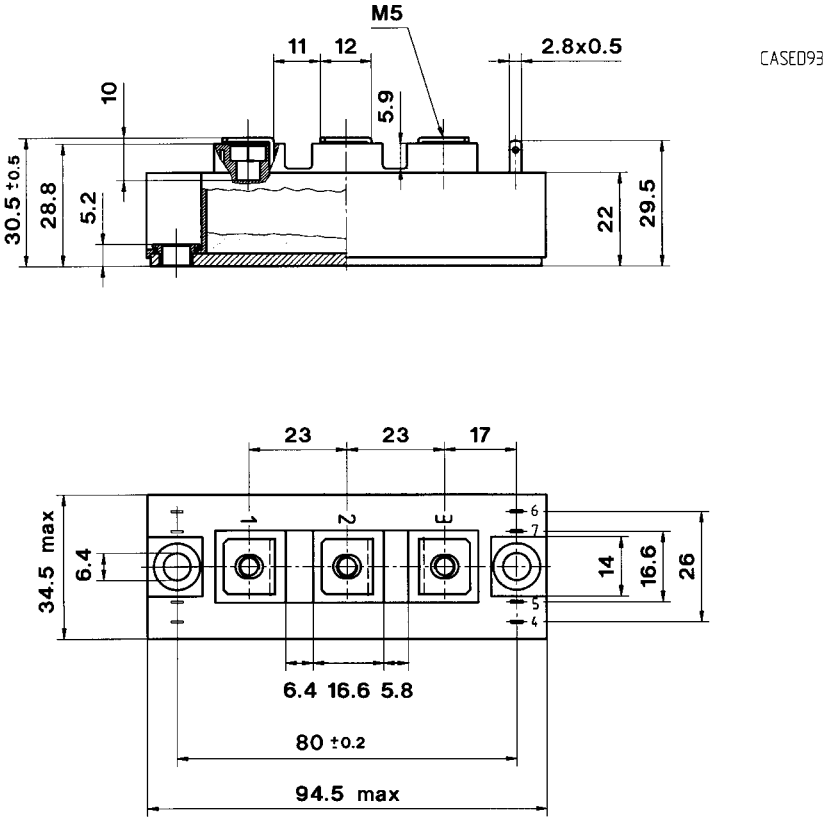
Fig. 24 Typ. CAL diode recovered charge

SEMITRANS 2N (low inductance)

Case D 93
UL Recognized
File no. E 63 532

SKM 195 GB 124 DN

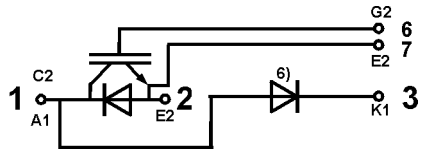




Dimensions in mm

SKM 195 GAL 124 DN

Case D 94 (→ D 93)



Case outline and circuit diagrams

Mechanical Data						
Symbol	Conditions		Values			Units
			min.	typ.	max.	
M_1	to heatsink, SI Units	(M6)	3	–	5	Nm
	to heatsink, US Units		27	–	44	lb.in.
M_2	for terminals, SI Units	(M5)	2,5	–	5	Nm
	for terminals, US Units		22	–	44	lb.in.
a			–	–	5x9,81	m/s^2
w			–	–	160	g

This is an electrostatic discharge sensitive device (ESDS). Please observe the international standard IEC 747-1, Chapter IX.

Twenty devices are supplied in one SEMIBOX D without mounting hardware, which can be ordered separately under Ident No. 33321100 (for 10 SEMITRANS 2)

Larger packing units of 20 pieces are used if suitable

This technical information specifies semiconductor devices but promises no characteristics. No warranty or guarantee expressed or implied is made regarding delivery, performance or suitability.

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